

In-situ JES: cut, timing and JER-smearing variations

γ +jet, p+p Run 24 data and Pythia8

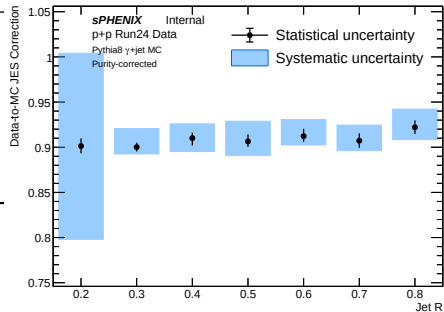
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Nominal in-situ JES (5 GeV jet cut)

R	p_a	stat $-/+$	syst $-/+$	previous
0.2	0.9013	0.0081 / 0.0085	0.103 / 0.103	0.9034
0.3	0.9001	0.0047 / 0.0049	0.008 / 0.021	0.9158
0.4	0.9101	0.0082 / 0.0061	0.015 / 0.016	0.9265
0.5	0.9064	0.0060 / 0.0078	0.015 / 0.022	0.9259
0.6	0.9123	0.0064 / 0.0081	0.010 / 0.018	0.9292
0.7	0.9072	0.0078 / 0.0081	0.011 / 0.017	0.9295
0.8	0.9220	0.0072 / 0.0077	0.014 / 0.020	0.9519

- ▶ Full pipeline on the Oct 5 trees; converged after 3 scans (last change 0.0005).
- ▶ “Previous” is the table from the older trees.
- ▶ R = 0.2 syst. is set by the three-jet variation, which sits at the scan edge (0.800).



Which jets are stored in the γ +jet trees

A jet is stored if any of these holds (CaloAna.cc):

- ▶ raw $p_T \geq 3$ GeV;
- ▶ calibrated $p_T \geq 5$ GeV, or calibrated / 0.90 ≥ 5 GeV;
- ▶ MC: any of the six smeared p_T values ≥ 5 GeV.
- ▶ Data: complete from calibrated 4.5 GeV. At 4.0–4.5 GeV only $\sim 20\%$ are stored.
- ▶ MC: the stored third-jet p_T (truth-anchored smear) is complete from 5.0 GeV.

Third-jet p_T , $R = 0.4$, entries per 0.25 GeV:

p_T (GeV)	Data	Pythia Photon10
3.75	23	40806
4.00	7860	41142
4.25	8973	41704
4.50	39860	41617
4.75	34049	41637
5.00	29493	68541
5.25	24796	63922

What the storage floor affects

Three-jet veto: data third jet $> 5 \text{ GeV} \times p_a$, with no $x_{J\gamma}$ floor.

- ▶ At $p_a = 0.80$ the veto is at 4.0 GeV, in the incomplete region, so data is under-vetoed.
- ▶ This is the variation that runs to the scan edge at $R = 0.2$.

Recoil jet: the scan applies the $x_{J\gamma}$ floor after dividing by p_a .

photon p_T	$x_{J\gamma}$ floor	softest calibrated jet used
15–20 GeV	0.40	$6.0 \times p_a$
20–25 GeV	0.30	$6.0 \times p_a$
25–35 GeV	0.30	$7.5 \times p_a$

- ▶ At the scan edge ($p_a = 0.80$) that is 4.8 GeV, above the 4.5 GeV completeness.
- ▶ The nominal p_a is not at a floor.

Treemaker fix (built on SDCC, needs a reprocess): store jets down to 4.0 GeV and add `thirdjet_pt_calib`.

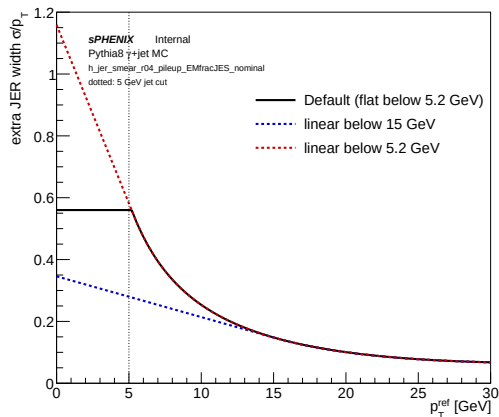
Jet p_T cut: 5, 7 and 9 GeV

Recoil-jet cut and three-jet veto changed together. 7 and 9 GeV: one scan, starting from the converged 5 GeV table.

R	0.2	0.3	0.4	0.5	0.6	0.7	0.8
5 GeV (nominal)	0.9013	0.9001	0.9101	0.9064	0.9123	0.9072	0.9220
7 GeV	0.8986	0.9070	0.9115	0.9140	0.9111	0.9067	0.9183
9 GeV	0.8895	0.9146	0.9168	0.9137	0.9047	0.8994	0.9127
7 – 5 GeV	−0.003	+0.007	+0.001	+0.008	−0.001	−0.000	−0.004
9 – 5 GeV	−0.012	+0.015	+0.007	+0.007	−0.008	−0.008	−0.009

- ▶ 7 GeV agrees with 5 GeV within 1% at every R.
- ▶ 9 GeV moves $R = 0.2$ by -1.3% and $R = 0.3$ by $+1.7\%$.
- ▶ The three-jet variation at $R = 0.2$ is off the scan edge: 0.9055 (7 GeV), 0.8982 (9 GeV).

JER smearing width below its measured range

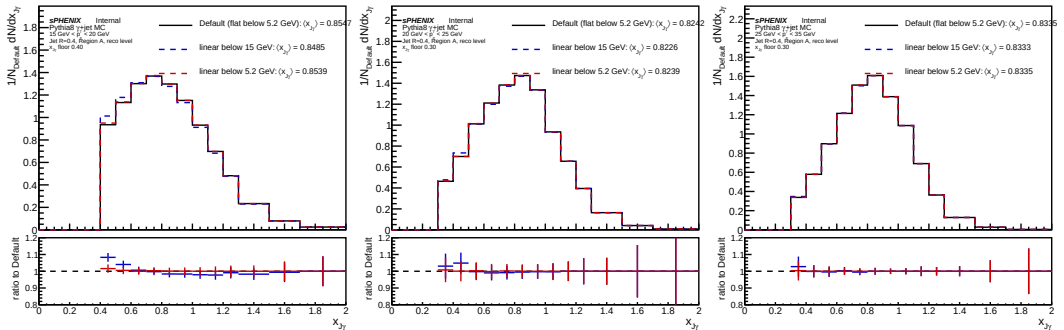


- ▶ Default: the template, flat below 5.2 GeV.
- ▶ Linear below 15 GeV: its tangent at 15 GeV.
- ▶ Linear below 5.2 GeV: its first segment.

Method (MC, nominal width only):

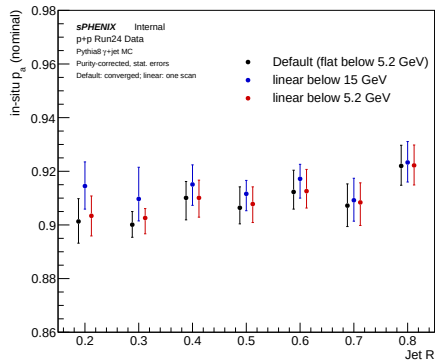
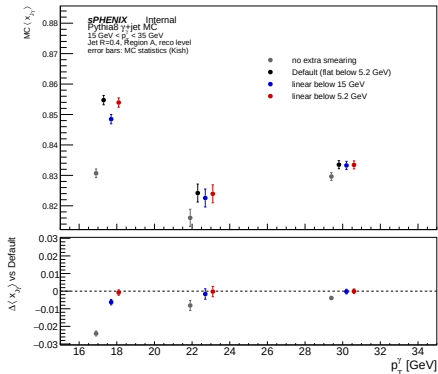
- ▶ Smear deviate recovered from the stored smeared p_T , then reused with the new width.
- ▶ Reference p_T : calibrated p_T below 5 GeV (exact), else the truth recoil jet within $0.75R$, else calibrated p_T .
- ▶ Region-A $R = 0.4$ pairs: 1.2% exact, 93.0% truth-matched, 5.8% fallback.

JER extrapolation: MC $x_{J\gamma}$, $R = 0.4$, region A



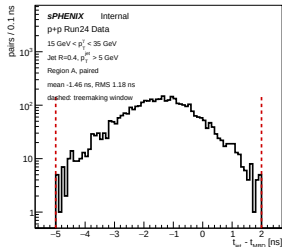
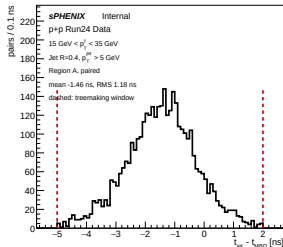
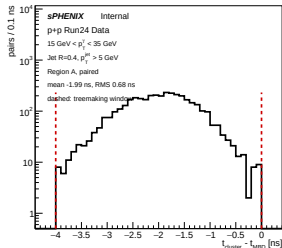
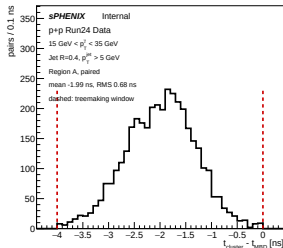
- ▶ All curves scaled by the Default histogram's integral; ratios to Default below.
- ▶ Linear below 15 GeV differs mainly at low $x_{J\gamma}$ in the 15–20 GeV bin.

JER extrapolation: MC reference and in-situ p_a



Δp_a vs Default	0.2	0.3	0.4	0.5	0.6	0.7	0.8
linear below 15 GeV	+0.013	+0.010	+0.005	+0.005	+0.005	+0.002	+0.001
linear below 5.2 GeV	+0.002	+0.003	0.000	+0.001	0.000	+0.001	0.000

Timing, $R = 0.4$: object time relative to the MBD

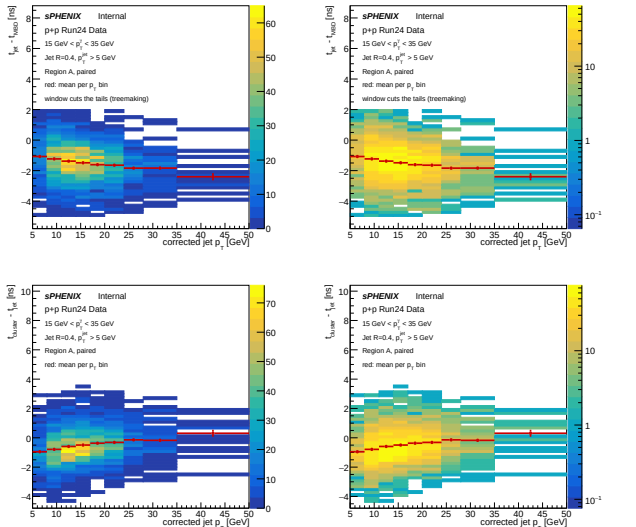


Region-A pairs in data, 15–35 GeV, 3729 pairs.

(ns)	mean	RMS
t_{cl}	-2.18	2.40
t_{jet}	-1.65	2.16
$t_{cl} - t_{MBD}$	-1.99	0.67
$t_{jet} - t_{MBD}$	-1.46	1.18
$t_{cl} - t_{jet}$	-0.53	1.12

- Treemaking windows (dashed):
cluster -4 to 0 ns, jet -5 to 2 ns.
- Stored objects are inside them.

Timing, $R = 0.4$: dependence on jet p_T



Mean per bin (red):

- ▶ $t_{jet} - t_{MBD}$: -1.08 ns at 5–8 GeV, -1.84 ns at 24–35 GeV.
- ▶ $t_{cl} - t_{jet}$: -0.95 ns at 5–8 GeV, -0.14 to -0.17 ns at 24–35 GeV.
- ▶ $t_{cl} - t_{MBD}$ vs photon p_T : flat at ~ -2.0 ns from 15 to 27 GeV.

Multijet timing cut vs γ +jet

Multijet (data, whole event dropped):

- ▶ Skim: $|t_{lead} + 2| < 6$ ns and $|t_{lead} - t_{sub}| < 3$ ns.
- ▶ Analysis: the same on $corrMaxJett/corrSubJett$, centred at 0.

γ +jet (data, per object, relative to MBD):

- ▶ Cluster: $0 < t_{MBD} - t_{cl} < 4$ ns.
- ▶ Jet: $-2 < t_{MBD} - t_{jet} < 5$ ns; a failing jet is replaced by the next one.

Multijet-style cuts on γ +jet region-A pairs (cluster as leading object):

cut	kept
$ t_{cl} - t_{jet} < 3$ ns	98.3%
$ t_{jet} + 2 < 6$ ns	99.4%
$ t_{cl} + 2 < 6$ ns	99.1%
skim-style: $ t_{cl} + 2 < 6, \Delta t < 3$	97.5%
analysis-style: $ t_{cl} < 6, \Delta t < 3$	92.8%

In-situ p_a with the timing cut, and with timing + 7 GeV

Data only: $|t_{cl} + 2| < 6$ ns and $|t_{cl} - t_{jet}| < 3$ ns. One scan each, from the converged 5 GeV table.

R	0.2	0.3	0.4	0.5	0.6	0.7	0.8
nominal	0.9013	0.9001	0.9101	0.9064	0.9123	0.9072	0.9220
timing cut	0.9034	0.9033	0.9158	0.9115	0.9172	0.9151	0.9290
timing + 7 GeV	0.9055	0.9199	0.9171	0.9168	0.9142	0.9104	0.9242
timing - nominal	+0.002	+0.003	+0.006	+0.005	+0.005	+0.008	+0.007
timing + 7 GeV - nominal	+0.004	+0.020	+0.007	+0.010	+0.002	+0.003	+0.002

- ▶ Timing cut: $R = 0.4$ region A 3832 $\rightarrow$ 3731 events. Every R moves up, within stat. (0.004–0.011).
- ▶ Timing + 7 GeV: $R = 0.4$ region A 3541 events. $R = 0.3$ moves by 2.2%, $R = 0.5$ by 1.1%, the rest within 0.7%.

Summary: change in nominal p_a

Δp_a vs nominal	0.2	0.3	0.4	0.5	0.6	0.7	0.8
7 GeV jet cut	-0.003	+0.007	+0.001	+0.008	-0.001	-0.000	-0.004
9 GeV jet cut	-0.012	+0.015	+0.007	+0.007	-0.008	-0.008	-0.009
JER linear below 15 GeV	+0.013	+0.010	+0.005	+0.005	+0.005	+0.002	+0.001
JER linear below 5.2 GeV	+0.002	+0.003	0.000	+0.001	0.000	+0.001	0.000
timing cut	+0.002	+0.003	+0.006	+0.005	+0.005	+0.008	+0.007
timing + 7 GeV	+0.004	+0.020	+0.007	+0.010	+0.002	+0.003	+0.002
nominal stat (-/+)	0.008/0.009	0.005/0.005	0.008/0.006	0.006/0.008	0.006/0.008	0.008/0.008	0.007/0.008

- ▶ Nominal p_a is 0.900–0.922 for all R; the variations are one scan each.
- ▶ The nominal scan is not limited by the tree storage floor; the three-jet veto at $R = 0.2$ is.